

MANISH MOHAN SINGH

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Education

Manipal University Jaipur

Aug 2025 – Jun 2027

Masters of Computer Applications

SGPA : 8.2

Graphic Era Deemed to be University

Aug 2022 – Jun 2025

Bachelor of Computer Applications

CGPA : 8.6

Projects

TaskForge | *React.js, Node.js, MongoDB, Redis, WebRTC, Socket.io, JWT, Nodemailer, Cloudinary*

Feb 2026

- Built an enterprise-grade SaaS platform for **project management**, **team collaboration**, **real-time communication**, and workflow **automation**.
- Optimized backend performance using MongoDB **aggregation** pipelines (*lookup, facet*) and Redis caching, reducing database queries by **98.6%** and achieving **sub-50ms** response times on cached requests.
- Implemented secure multi-device authentication with JWT rotation, HttpOnly cookies, token blacklisting, **rate limiting**, and **automated email notifications** via **cron jobs** and Redis-based deduplication.

NuraChat | *React, Node.js, Express, MongoDB, Passport.js, LiveKit, Hugging Face, Zustand, Tailwind CSS*

April 2026

- Engineered a secure real-time messaging platform with **AES-256-GCM** encrypted communication, WebRTC calling, and **AI-powered** multilingual translation.
- Developed an AI translation pipeline using mBART-50 and SHA-256 content-addressed caching to reduce translation latency and API costs
- Built secure authentication workflows with **OTP verification**, bcrypt hashing, MongoDB TTL expiration, and **OAuth** integration..

VideoChat — Open Source Contribution | *React, JavaScript, Expressjs, MongoDB, Nodejs*

Apr 2025

- Contributed to an open-source video conferencing platform by implementing advanced search and filtering capabilities, improving content discoverability and user navigation.
- **Optimized** backend **performance** using MongoDB **aggregation pipelines**, and query refactoring, reducing memory overhead and database response times.
- Enhanced database **scalability** through strategic **indexing**, improving query efficiency for frequently accessed collections and search operations.

Technical Skills

Tech Stack: React.js, Node.js, Express.js, MongoDB, Redis, Socket.IO, WebRTC, LiveKit, Tailwind CSS, JWT, Cloudinary

Languages: JavaScript, TypeScript, Java, HTML, CSS, SQL

Developer Tools: Git, GitHub, Postman, VS Code, MongoDB Atlas, Vercel, Render, Linux

Technologies: Git, Linux, Docker, Kubernetes,

Certifications

The Complete Full-Stack Web Development Bootcamp — Angela Yu

Jul 2025

Udemy

- Understood the roles and responsibilities of Front-End, Back-End, and Full-Stack Developers.
- Designed and integrated RESTful APIs to enable communication between frontend and backend systems.
- Worked with user authentication, MongoDB and PostgreSQL for data modeling, CRUD operations, and persistence
- Understood client-server architecture, HTTP protocol, and web application request-response lifecycle.

Achievements

- Solved 300+ Data Structures and Algorithms problems across LeetCode, academic coursework, and coding practice platforms, covering Arrays, Strings, Linked Lists, Trees, Graphs, Dynamic Programming, and Greedy Algorithms.
- Completed industry-focused software engineering simulation with **Accenture**, gaining experience in code review, debugging, testing, and Agile workflows
- Developed and deployed multiple full-stack web applications using React, Node.js, Express.js, MongoDB, and PostgreSQL, demonstrating proficiency in modern web development and API-driven architectures.